

Manav R Gurnani

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EDUCATION

University of California, Davis

B.S. in Computer Science & Statistics (ML Concentration)

Sep 2023 - Aug 2026

Davis, CA

EXPERIENCE

Amazon Web Services (AWS)

Software Development Engineer Intern (AWS Amplify)

Jun - Sep 2025

Seattle, WA

- Reduced backend configuration and CI/CD efforts by 65% by architecting an automated CloudFormation IaC framework within AWS Amplify, eliminating redundancy in Geo and RESTful configurations
- Improved platform reliability for 2,100+ customers by building reusable TypeScript CDK components that abstracted resource orchestration and service integration
- Lowered REST API onboarding overhead by designing API Gateway routing and Cognito authorization abstractions, standardizing authentication and request flows across Amplify applications
- Increased API development velocity by 75% by introducing Express-style Lambda backends into Amplify's REST service

EcoCAR at UC Davis

Undergraduate Software Lead - Connected Autonomous Vehicle (CAV)

Jan 2024 - Jun 2026

Davis, CA

- Led a 15-member autonomous systems team that designs perception, fusion, and driver-assistance features for an industry-level connected autonomous vehicle
- Improved object detection and tracking accuracy by 50% by implementing a C++ sensor-fusion pipeline combining computer vision algorithms, k-means clustering for object classification, and Kalman filter-based multi-target tracking
- Developed a real-time driver monitoring system through signal processing and time-series statistical inference to detect behavioral anomalies and trigger driver-attention alerts

VectorEdge

Software Engineering Intern (Data Loss Prevention)

Jun - Aug 2024

San Jose, CA

- Built a production-style CLI agent supporting user authentication, query retrieval, and recursive scanning for 2,000+ users
- Reduced search latency by 85% by implementing a search engine with Aho-Corasick automata and SQLite indexing
- Designed a cross-language gRPC service with a C++ client and Go backend communicating for secure query execution

RESEARCH

Aviran Lab

Undergraduate Researcher

Jan 2026 - Present

Davis, CA

- Investigated multimodal feature integration across architecture components (GraphConv, cross-attention, transformer, DPRBP) in a state-of-the-art RNA-protein interaction model (PaRPI)
- Extending BRIDGE, an existing motif-enhanced RNA-binding predictor with protein-aware context via cross-attention, investigating improvement in binding prediction generalization across protein and cell types
- Improved computational efficiency by 20% by incorporating multi-threading throughout embedding and model extraction pipelines

Karzand Lab

Undergraduate Researcher

Jun 2026 - Present

Davis, CA

- Investigated whether heuristic information-theoretic proxies in the InfoBoost transfer-learning algorithm accurately approximate true mutual information and KL divergence, using closed-form Gaussian mean-estimation experiments to test ranking fidelity across sample size and domain shift

Real-time Monocular Lane Segmentation and Vehicle Tracking

Undergraduate Researcher - UC Davis EcoCAR Senior Design

Jan 2026 - Jun 2026

Davis, CA

- Reduced autonomous sensory requirements by 85% by engineering a monocular ML-based vehicle perception system with object segmentation, tracking, and lane detection
- Optimized real-time inference pipelines by implementing a monolithic transformer framework for perception tasks and a continual learning pipeline for refined decision-making improvements

TECHNICAL SKILLS

Languages: Java, Python, C/C++, TypeScript, SQL, JavaScript, Go, HTML/CSS, Intel x86 Assembly, XML, R

Technologies: TensorRT, PyTorch, AWS CDK, AWS CloudFormation, AWS Lambda, Amazon Cognito, OpenCV, RTMaps, React.js, Docker, MongoDB, Node.js, Express.js, MySQL CMake, Github, Snowflake, Slurm Scripting, MATLAB

Concepts: Data Structures and Algorithms, Embedded Systems, REST APIs, Cloud Infrastructure-as-Code (IaC), Serverless Architecture, Probability Theory, Information Theory, High Performance Computing, Real-time ML Inference, Bayesian Learning, Continual Learning, Transformers, CNNs